SELF ASSESSMENT 1-2 (Page 28)

1. $P_1 = 500\ \text{kPa} \qquad V_1 = 0.1\ \text{m}^3 \quad P_2 = 100\ \text{kPa}$

$P_1V_1 = P_2V_2$

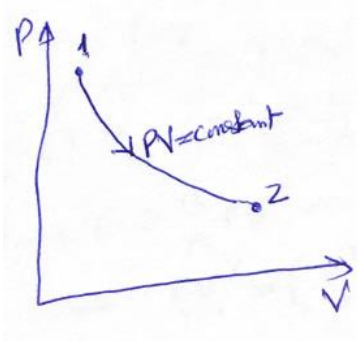
$V_2 = \frac{P_1}{P_2} \times V_1 = \frac{500}{100} \times 0.1\text{m}^3 = 0.5\ \text{m}^3$

$\text{Workdone } W_{12} = -\int PdV = -\int_1^2 \frac{C}{V}dV = -P_1V_1 \text{In}\left(\frac{V_2}{V_1}\right)$

$W_{12} = -500 \times 0.1\text{In}\left(\frac{0.5}{0.1}\right) = -80.47$

$W_{12} = \mathbf{-80.5\ kJ}$

The work done by the gas is 80.5 kJ



TYPICAL UNIT 2 CALCULATION MCQ'S IN ME 265 - DETAILED SOLUTIONS

1. Work done in a polytropic process is

$W = \int_1^2 PdV \qquad \text{but } PV^n = C \text{ and } P = CV^{-n}$

$\text{hence } W = \int_1^2 V^{-n}dV = C\frac{[V_2^{1-n} - V_1^{1-n}]}{1-n} = \frac{P_2V_2 - P_1V_1}{1-n}, n \neq 1$

also for a gas $PV = mRT$

At state 1 $\frac{P_1V_1}{T_1} = mR;\ mR = \frac{300\times0.2}{373.15} = 0.1608\ \text{kJ/K}$

$\text{Hence, } W = \frac{mR(T_2 - T_1)}{1-n} = \frac{0.1608(473.15 - 373.15)}{1-1.2} = \mathbf{-80.4\ kJ}$

2. $Q_{12} = 0 \text{ and } S_2 = S_1 \text{(Reversible adiabatic process);} \qquad \text{for Air } \gamma = 1.4$

$\text{Hence, } T_2 = T_1\left(\frac{P_2}{P_1}\right)^{\gamma-1/\gamma} = 100\left(\frac{100}{1500}\right)^{0.4/1.4} = 461.3\ \text{K}$

$Q_{12} - W_{12} = \Delta E = \Delta U = mC_v(T_2 - T_1)$

$W_{12} = -mC_v(T_2 - T_1) = mC_v(T_1 - T_2) = 1 \times 0.717 \times (1000 - 461.3) = \mathbf{386.2\ kJ}$

3. $PV^{1.15} = \text{constant}$

$T_2 = T_1\left(\frac{P_2}{P_1}\right)^{n-1/n} = 293.15\left(\frac{800}{100}\right)^{0.15/1.5} = 384.5\ \text{K}$

$W_{12} = \frac{mR(T_2 - T_1)}{1-n} = \frac{1 \times 0.51835(384.5 - 293.15)}{1-1.15} = \mathbf{-315.7\ kJ}$

4. $PV^{1.5} = \text{constant};\ V_2/V_1 = 8$

State 1: $P_1 = 7\ \text{Mpa}, T_1 = 1800\ \text{K}, V_1 = 0.2\ \text{L},$

$m_1 = P_1V_1/RT_1 = \frac{7 \times 10^3 \times 0.2 \times 10^{-3}}{0.287 \times 1800} = 2.71 \times 10^{-3}\ \text{kg}$

State 2: $T_2 = T_1\left(\frac{V_1}{V_2}\right)^{n-1} = 1800\left(\frac{1}{8}\right)^{0.5} = 636.4\ \text{K}$

$W_{12} = \frac{mR(T_2 - T_1)}{1-n} = \frac{2.71 \times 10^{-3} \times 0.287(636.4 - 1800)}{(1-(1.5))} = \mathbf{1.81\ kJ}$

5. $m_1 = {}^{P_1V_1}/_{RT_1} = \frac{110\times0.1}{0.287\times298.15} = 0.1286\ \text{kg}$

$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{{}^{n-1}/_n} = \frac{473.15}{298.15} = \left(\frac{800}{110}\right)^{{}^{n-1}/_n}$

${}^{n-1}/_n = \frac{\log(473.15/298.15)}{\log(800/100)} = \frac{0.200564}{0.861697} = 0.2328$

Hence, $n = 1.3034$

$V_2 = V_1\left(\frac{P_2}{P_1}\right)^{{}^1/_n} = 0.1\left(\frac{110}{800}\right)^{0.7672} = 0.02182\ \text{m}^3$

$W_{12} = \frac{P_2V_2 - P_1V_1}{1-n} = \frac{[800 \times 0.02182 - 110 \times 0.1]}{(1-1.3034)} = \mathbf{-21.28\ kJ}$

6. $W_{12} = \frac{P_2V_2 - P_1V_1}{1-n} = \frac{mR(T_2 - T_1)}{1-n}$

Specific work $= \frac{W_{12}}{m} = \frac{R(T_2 - T_1)}{1-n} = \frac{0.287 \times (500 - 300)}{1 - 1.3} = \mathbf{-191.3\ kJ}$

7. $PV = C$ and $P = CV^{-1}$; $P_2V_2 = P_1V_1$

$V_2 = \frac{P_1V_1}{P_2} = \frac{1000 \times 0.1}{100} = 1\ \text{m}^3$

$W = \int_1^2 PdV = \int_1^2 C\frac{dV}{V} = C\text{In}\left(\frac{V_2}{V_1}\right) = 1000 \times 0.1\text{In}\left(\frac{1}{0.1}\right) = \mathbf{230.3\ kJ}$

8. $PV^n = \text{constant}$

$T_2 = T_1 = \left({}^{P_2V_2}/_{P_1V_1}\right) = T_1\left(\frac{P_2}{P_1}\right) \times \left(\frac{P_2}{P_1}\right)^{{}^1/_n} = 400\left(\frac{300}{100}\right)^{\frac{1.3-1}{1.3}} = 515.4\ \text{K}$

$\text{Work} = \frac{P_2V_2 - P_1V_1}{1-n} = \frac{mR(T_2 - T_1)}{1-n} = \frac{0.1 \times 0.287(515.4 - 400)}{1 - 1.3} = \mathbf{-11.04\ kJ}$

9. $PV^{1.5} = \text{constant}$

State 1: $v_1 = 0.09921\ \text{m}^3/\text{kg}$ $\quad P = P_{sat} = 201.7\ \text{kPa}$

State 2: $V_2 = V_1\left(\frac{P_2}{P_1}\right)^{{}^1/_{1.5}} = 0.09921 \times \left(\frac{201.7}{500}\right)^{\frac{1}{1.5}} = \frac{0.05416\text{m}^3}{\text{kg}}$

State 2 is superheated, $T_2 = 79\ ^\circ\text{C}$

$W_{12} = \frac{m(P_2V_2 - P_1V_1)}{1-1.5} = \frac{0.5(500 \times 0.5416 - 201.7 \times 0.09921)}{-0.5} = \mathbf{-7.07\ kJ}$

10. $PV^n = \text{constant}$

$\left(\frac{P_2}{P_1}\right) = \left(\frac{V_2}{V_1}\right)^n$

$$n = In\left(\frac{P_2}{P_1}\right)/In\left(\frac{V_2}{V_1}\right) = \frac{1.0986}{0.57246} = 1.919$$

$$\text{Work} = \frac{P_2V_2 - P_1V_1}{1 - n} = \frac{(1000 \times 0.20596 - 3000 \times 0.11619)}{1 - 1.919} = \mathbf{155.2\ kJ}$$